PHYS4055 Week 3 Tutorial Notes

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I. 2024 EXAM SOLUTION

A. Question 3

(10 marks)

A beam-splitter is an optical device with two input modes and two output modes. Consider a 50:50 beam splitter with input modes 1 and 2 and output modes 3 and 4. This beam-splitter combines the modes as $\hat{a}_1 = (\hat{a}_3 + \hat{a}_4)/\sqrt{2}$ and $\hat{a}_2 = (\hat{a}_3 - \hat{a}_4)/\sqrt{2}$.

1. (3 marks) What output state is expected when the two inputs each have a single photon in them $|1\rangle_1|1\rangle_2$?

$$\hat{a}_1^\dagger \hat{a}_2^\dagger |0\rangle_1|0\rangle_2 \to \frac{1}{2}\left(\hat{a}_3^\dagger + \hat{a}_4^\dagger\right)\left(\hat{a}_3^\dagger - \hat{a}_4^\dagger\right)|0\rangle_3|0\rangle_4 = \frac{1}{\sqrt{2}}\left(|2\rangle_3|0\rangle_4 - |0\rangle_3|2\rangle_4\right) \tag{1}$$

2. (2 marks) Show that for any input this beam-splitter conserves energy.

$$\begin{aligned} E &\propto \sum_m \hat{n}_m \\ \hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2 &= \left(\frac{\hat{a}_3^\dagger + \hat{a}_4^\dagger}{\sqrt{2}}\right)\left(\frac{\hat{a}_3 + \hat{a}_4}{\sqrt{2}}\right) + \left(\frac{\hat{a}_3^\dagger - \hat{a}_4^\dagger}{\sqrt{2}}\right)\left(\frac{\hat{a}_3 - \hat{a}_4}{\sqrt{2}}\right) \\ &= \hat{a}_3^\dagger\hat{a}_3 + \hat{a}_4^\dagger\hat{a}_4 \end{aligned} \tag{2}$$

Consider the normalised state $|\psi\rangle = \alpha|n\rangle + \beta|n+1\rangle$ where $|n\rangle$ is the n-photon state and $|\alpha|^2 + |\beta|^2 = 1$.

3. (3 marks) Show that this state exhibits sub-Poissonian statistics, i.e. $\mathrm{Var}(n) < \langle n\rangle$.

$$\begin{aligned} \mathrm{Var}(n) &= \langle\hat{n}^2\rangle - \langle\hat{n}\rangle^2 \\ \langle\hat{n}^2\rangle &= |\alpha|^2 n^2 + |\beta|^2(n+1)^2 = n^2 + |\beta|^2(2n+1) \\ \langle\hat{n}\rangle &= |\alpha|^2 n + |\beta|^2(n+1) = n + |\beta|^2 \\ \mathrm{Var}(n) &= n^2 + |\beta|^2(2n+1) - (n+|\beta|^2)^2 = n^2 + |\beta|^2(2n+1) - n^2 - 2n|\beta|^2 - |\beta|^4 \\ &= |\beta|^2 - |\beta|^4 \end{aligned} \tag{3}$$

$$\mathrm{Var}(n) = |\beta|^2 - |\beta|^4 < |\beta|^2 + n = \langle\hat{n}\rangle \qquad n \geq 0 \tag{4}$$

4. (2 marks) Briefly describe the significance of sub-Poissonian statistics in light.
No semi-classical explanation of sub-Poissonian statistics, demonstrates that light is quantized.

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